

# Urban-Rural Stratification for Improved NTL-Based Population Proxy Estimation Using VIIRS Data

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## Abstract.

We propose a data-driven urban-rural stratification method using KMeans clustering ( $k=2$ ) on district-level VIIRS night-time light (NTL) features to improve population estimation. The method requires no census or land-cover inputs. Evaluated on 670 Indian districts and 695 Nigerian Local Government Areas with 5-fold cross-validation, stratification raises  $R^2$  from 0.829 to 0.858 in India ( $p=0.027$ ) and from 0.168 to 0.251 in Nigeria ( $p=0.004$ , +50% relative gain). Ablation experiments confirm that NTL density normalisation is counter-productive, NDVI correction fails in tropical regions, and  $k=2$  is optimal. All code and data are openly available.

## 1. INTRODUCTION

Sub-national population estimates are needed for resource allocation, disaster response, and Sustainable Development Goal monitoring. Night-time light (NTL) imagery from VIIRS is a low-cost, globally available proxy, but standard approaches fit a single regression across all administrative units. This ignores the fundamental structural difference between urban districts (where NTL saturates at high population) and rural districts (where NTL-population coupling is weaker). We propose a simple stratification: KMeans on NTL features separates urban and rural units, and independent models are fitted for each cluster.

## 2. DATA AND METHODOLOGY

**Study areas.** India (670 districts, high electrification) and Nigeria (695 LGAs, lower electrification, gas flaring).

**Data.** VIIRS VNL V2.1 annual composites (2019), WorldPop UN-adjusted population (1 km), GADM v4.1 boundaries.

**Features.** Five district-level features via zonal statistics: `ntl_sum`, `ntl_mean`, `ntl_max`, `lit_fraction`, `log_area`, all  $\log(1+x)$  transformed.

**Pipeline.** KMeans ( $k=2$ ) clusters districts into urban/rural groups using only NTL features (Fig. 1). Independent log-linear regressions are fitted per cluster. Evaluation: 5-fold CV with paired  $t$ -tests.

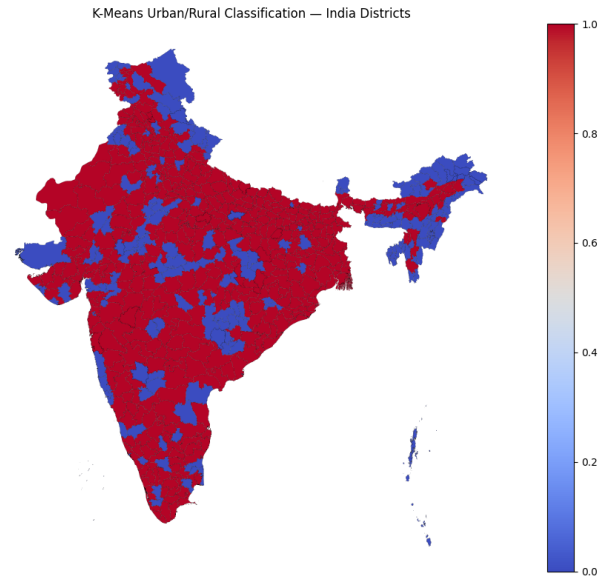
## 3. RESULTS

### 3.1 NTL-Population Relationship

Fig. 2 shows that the raw NTL-population distribution is heavily right-skewed; after log transformation, the correlation reaches  $r=0.822$ , validating the log-linear framework. Spread at high NTL values reflects urban sensor saturation.

### 3.2 Stratification Performance

Table 1 shows statistically significant gains in both countries. Nigeria’s lower baseline  $R^2$  is due to gas flares, La-



**Figure 1:** KMeans ( $k=2$ ) urban-rural classification of 670 Indian districts using NTL features only. Cluster 0: 464 urban, Cluster 1: 206 rural.

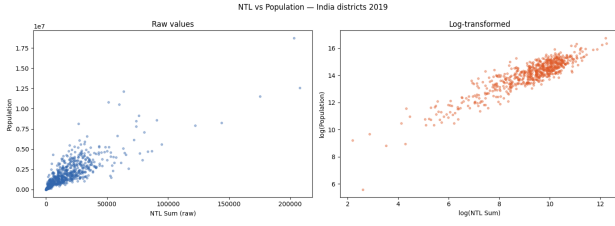
**Table 1:** Cross-validated  $R^2$  (5-fold, paired  $t$ -test).

|         | $n$ | Base  | Strat.       | $p$   |
|---------|-----|-------|--------------|-------|
| India   | 670 | 0.829 | <b>0.858</b> | 0.027 |
| Nigeria | 695 | 0.168 | <b>0.251</b> | 0.004 |

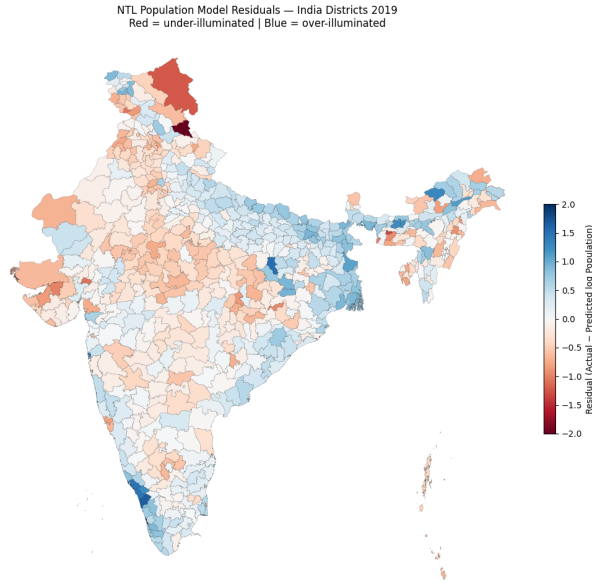
gos sensor saturation, and low northern electrification, but stratification delivers the largest *relative* gain (+50%).

### 3.3 Ablation Studies

- District area is not a confound ( $r=0.119$ ).
- NTL density normalisation drops  $R^2$  from 0.83 to 0.47.
- NDVI correction fails in Kerala (dense vegetation + dense population).
- $k=2$  outperforms median split (silhouette 0.54 vs. 0.51);  $k=3$  adds only +0.002 ( $p=0.41$ ).



**Figure 2:** NTL vs. population for India. Left: raw (power-law). Right: log-transformed ( $r=0.822$ ).



**Figure 3:** Residual map for India. Red: under-predicted, Blue: over-predicted. Systematic failures in Kerala and Mumbai.

### 3.4 Residual Analysis

Fig. 3 maps prediction residuals. The largest under-predictions are in Kerala (dispersed settlement, low commercial NTL) and Mumbai Suburban (informal settlements), revealing systematic NTL failure modes.

## 4. SCIENTIFIC CONTRIBUTION & CONCLUSIONS

The primary contribution of this work is proving that an unsupervised, fully data-driven KMeans stratification on NTL features significantly improves population proxy estimation across two structurally diverse countries. Crucially, the method requires *no* auxiliary census or land-cover data, making it directly and immediately transferable to any developing nation with global VIIRS coverage.

By separating the analysis into India (where the method pushes  $R^2$  to 0.858) and Nigeria, we demonstrate both the peak capability and the boundary conditions of NTL proxies. Nigeria’s low absolute  $R^2$  is framed not as a failure, but as a critical diagnostic finding: it identifies exactly where and why NTL-based models break down (gas flares, low rural electrification, and urban sensor saturation). Despite these extreme conditions, stratification still yielded a +50% relative performance gain, proving that regime-

specific modelling is most vital exactly where heterogeneity is highest. Future work will integrate dedicated gas flare masking, multi-year temporal composites, and auxiliary open-source features (e.g., OpenStreetMap building footprints) to further refine these estimates in challenging topographies.

**Code and data:** <https://github.com/light-oishikk/NTL-Population-Proxy>

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